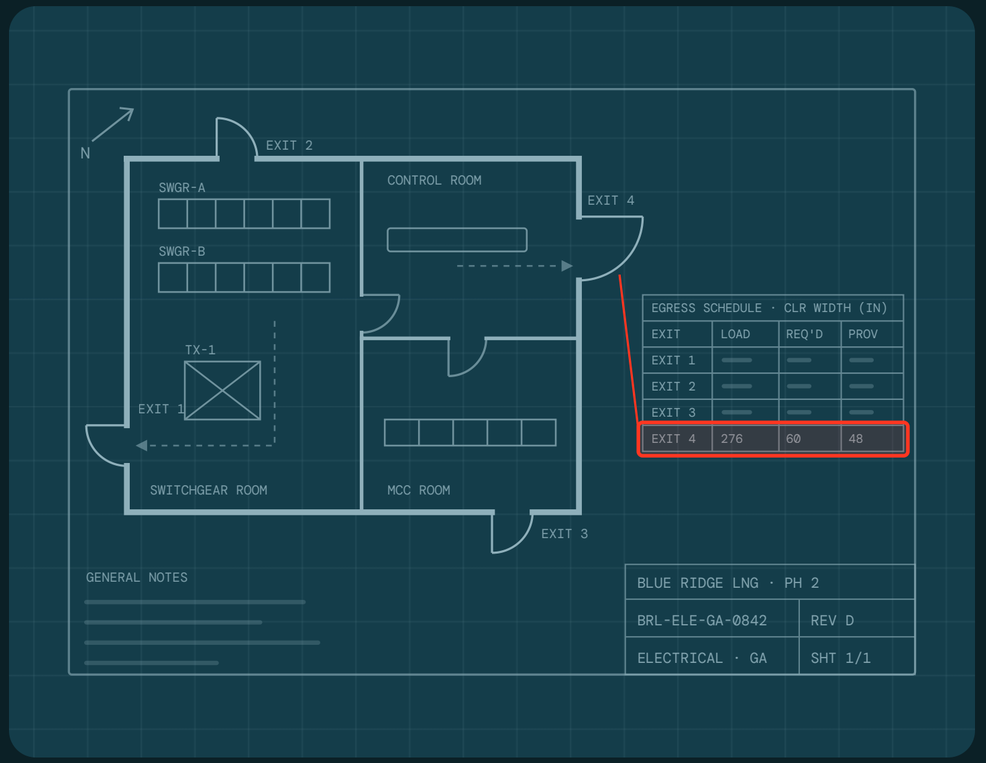
databricks

DATABRICKS × BECHTEL

AI-assisted project intelligence for mega-projects

Automated drawing compliance and a PM Co-Pilot on Databricks



November 2026

THE PROBLEM

Manual drawing audits are slow, and violations found late become rework and slip

Thousands of drawings, checked by hand

Every isometric, P&ID and general arrangement is audited against code, on billable engineering hours.

Late findings cost the most

A violation caught after issue means rework, schedule slip and safety exposure.

The project picture is split across tools

Schedule, cost, safety, quality and compliance sit apart. Each question becomes an analyst request.

Checked against codes such as `ASME B16.5` `ASME B31.3` `OSHA 1910.147` `IBC 1005.1`

WHAT A PROJECT MANAGER NEEDS DAY-TO-DAY OWNER

One place to ask, plain-English answers, and evidence they can trust

TODAY		WHAT THE PM NEEDS
Schedule, cost, safety, quality and compliance live in separate tools	→	**One place** The whole project on one screen, every morning
A question becomes a request to an analyst, then a wait	→	**Plain-English answers** Ask directly; get the answer, the table and the query behind it
Findings arrive without the drawing or the clause attached	→	**Evidence they can trust** Every number checked; every finding linked to its drawing and code clause

Every answer comes with its evidence

1 Ask in plain English 2 See the answer, the SQL and the result 3 Open the drawing at the violation, clause explained

Project Co-Pilot Blue Ridge LNG Terminal · Phase 2

Prototype UI · synthetic project data PM

Schedule	Budget	Safety	Open NCRs	Schedule-slip risk · next 90 days
On track · 6 of 11 key milestones	+2.3% EAC · $4.8M over	0 TRIR · 28 days incident-free	24 · 9 compliance-related	26 · High risk milestones

Daily brief

Numbers checked

- 51 of 238 drawings flagged for engineer review this week
- 6 milestones are already late and 1 is due today
- 24 NCRs open; 9 compliance-related
- 26 milestones are High risk in the next 90 days

Late or due today

MS-281	Cryogenic tank T-203	**4 days late**
MS-271	Heat exchanger install	**15 days late**
MS-277	Align equip., Train 3	**11 days late**

1 **Which drawings failed this week and why?**

51 of 238 drawings reviewed this week were flagged for engineer review, with 85 findings in total.

2 **Generated SQL** Genie · runs as the signed-in user

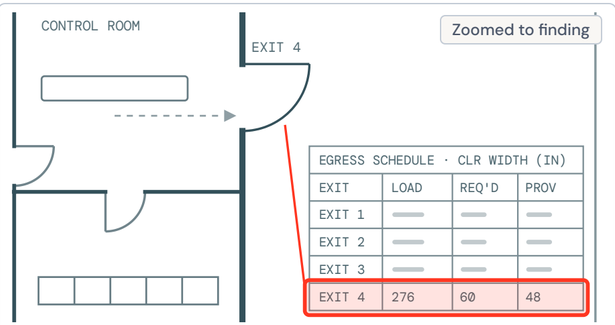
```
SELECT drawing_id, discipline, citation, reason
FROM   bluerridge_gold_pm.compliance_findings
WHERE  status = 'FAIL' AND review_date BETWEEN '2026-10-31' AND '2026-11-06'
ORDER BY drawing_id;
```

Drawing	Discipline	Citation	Reason
BRL-ELE-GA-0842	Electrical	IBC 1005.1	Exit 4 clear width 48 in; 60 in required
BRL-ELE-GA-0844	Electrical	OSHA 1910.147	LOTO tag format mismatch, LCP-3805
BRL-ELE-GA-1087	Electrical	IBC 1005.1	Exit 3 clear width 36 in; 44 in required

Showing 3 of 85 findings on 51 drawings

Sources BRL-ELE-GA-0842.pdf BRL-ELE-GA-0844.pdf BRL-ELE-GA-1087.pdf

3 **BRL-ELE-GA-0842.pdf** Rev D · Electrical · page 1



IBC 1005.1 · Minimum required egress width

Egress width must cover the occupant load served. Exit 4 serves 276 occupants and needs 60 in; the drawing provides 48 in.

Evidence `EXIT 4 … REQ'D CLR WIDTH (IN) 60 PROVIDED (IN) 48`

⚠ AI finding · verify before action

FROM DRAWING TO DECISION

Every drawing is checked automatically, and engineers review the flagged ones first

1

Read

Parse every drawing. Capture title-block fields and the drawing type.

AI Functions

`ai_parse_document`
`ai_extract · ai_classify`

EXAMPLE

`BRL-ELE-GA-0842`
GA · Electrical · Rev D

2

Check

An LLM judge tests it against the rule set. Deterministic guardrails confirm.

AI Functions

`ai_query + guardrails`

EXAMPLE

Flagged
IBC 1005.1 egress width

3

Explain

The finding cites the code clause, quotes the evidence and says why.

Vector Search

`code-clause summaries`

EXAMPLE

Exit 4 gives 48 in; 276 occupants need 60 in

4

Triage

Engineers work one queue of flagged drawings, opened at the violation.

Databricks Apps

`engineer in the loop`

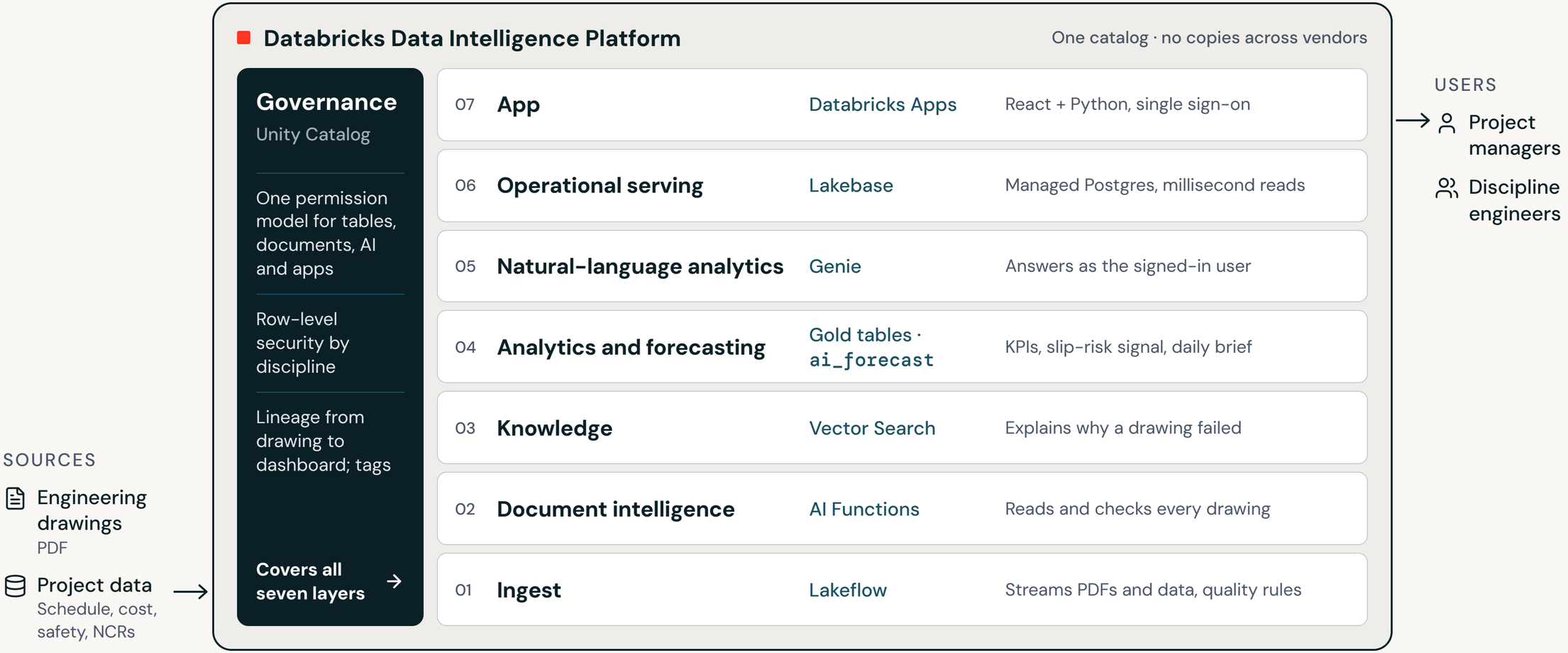
PROTOTYPE WEEK

51 of 238 drawings to review

Every finding lands as a governed table in Unity Catalog, ready for KPIs, Genie and the app.

HOW IT WORKS

Every layer runs inside one governed platform



PROTOTYPE RESULTS

The prototype narrowed a week's 238 drawings to 51 for engineer review

1,200

drawings read automatically

0 parse errors. Drawing type matched the reference 100%; discipline 99.8%.

51 of 238

drawings flagged for engineer review in the week reviewed

~81%

of planted code violations caught

Recall 81.0% · precision 58.1%

15 of 15

benchmark PM questions answered correctly by Genie

Including 5 it had never seen

~3 ms

app data response from Lakebase

vs ~430 ms over a warehouse query

A triage assistant, with an engineer in the loop

About 58% of flags were real violations, so an engineer reviews every flag. Next: tune on Bechtel's own standards and historical NCRs.

Prototype built on representative synthetic project data; ready to pilot on a live Bechtel project. Fictional Blue Ridge LNG Terminal, Phase 2: 1,200 synthetic drawings plus synthetic schedule, cost, safety and NCR data. Prototype measurements, not production guarantees.

THE INVESTMENT CASE **EXECUTIVE SPONSOR**

The same results, in the executive sponsor's terms

WHAT THE PROJECT MANAGER SEES Runs the project day to day		**WHAT THE EXECUTIVE SPONSOR GETS** Funds the portfolio rollout: Chief Project Officer, VP Engineering
51 of 238 drawings flagged this week, each opened at the violation	→	**Rework avoided** Design errors caught on the drawing, before they become field rework
26 High-risk milestones in the next 90 days; **6** already late	→	**Risk exposure reduced** Compliance, schedule and safety exposure seen weeks earlier
Plain-English answers in seconds, with the SQL and the source drawing	→	**A measured return before scaling** A 6–8 week pilot with a go/no-go gate before portfolio spend

ⓘ Next: what that rework is worth, and what it takes to pay back the pilot. Public benchmarks stand in until the pilot measures Bechtel's own numbers.

THE INVESTMENT CASE **EXECUTIVE SPONSOR**

Catching design errors early is worth about $3M per $1B of construction

CONSTRUCTION SPEND
$1B
Per $1B, so it scales to any project or portfolio

×

DIRECT REWORK
5%
Industry benchmark: 4–5% of construction cost[1]

×

SHARE VISIBLE ON DRAWINGS
30%
Assumed. Design-induced rework is over 70% of the total[2]

×

CAUGHT EARLY BY THE CO-PILOT
20%
Assumed; the pilot measures it. The prototype caught 81% of planted violations

=

REWORK AVOIDED
$3.0M
Per $1B of construction, direct cost only

SCENARIO	SHARE VISIBLE ON DRAWINGS	CAUGHT EARLY	AVOIDED PER $1B	ON A $10B ANNUAL PORTFOLIO
Conservative	20%	10%	$1.0M	$10M a year
Base	**30%**	**20%**	**$3.0M**	**$30M a year**
Upside	40%	30%	$6.0M	$60M a year

An illustrative model built on public benchmarks, not a measured result. Direct rework only; delay, claims and indirect costs are excluded. The $10B portfolio is an example input; the pilot replaces each assumption with Bechtel's data. [1] Construction Industry Institute, RS 203-1 (2005); Hwang et al., *J. Constr. Eng. Manage.* 135(3), 2009. [2] Love, Edwards and Irani, *IEEE Trans. Eng. Manage.* 55(2), 2008.

THE INVESTMENT CASE **EXECUTIVE SPONSOR**

The pilot pays for itself at a ~3% catch rate on one $1B project

$0.5M

pilot budget ceiling, all-in

Assumed, and set high on purpose: 6–8 weeks on one project, 2–3 Bechtel SMEs, platform and delivery support

~3%

of one $1B project's drawing-visible rework repays the pilot

$0.5M ÷ $15M ($1B × 5% rework × 30% visible). The base case assumes 20% is caught early

$30M

a year in avoided rework at portfolio scale

Base case on a $10B annual portfolio. Range $10M–$60M, before run-rate platform cost

RISK EXPOSURE REDUCED

Compliance
Code violations found at design review, not at inspection or in the field

Schedule
Milestone slip risk ranked 90 days ahead. Large projects typically finish 20% late[1]

Safety
Lockout/tagout and egress issues flagged before field work begins

The week-8 readout tests this case with measured numbers. Pilot cost, catch rate and run-rate cost are measured on Bechtel's project, and the go/no-go decision on a portfolio rollout rests on them. [1] McKinsey, "Imagining construction's digital future" (2016).

TRUST AND GOVERNANCE

Every answer is governed, shows its work, and leaves the decision to an engineer

Governed data

Unity Catalog

One permission model

Across tables, documents, AI and apps

Row-level security by discipline

Unity Catalog row filters restrict each user's view; demonstrated on the drawing register through Genie

Genie answers as the signed-in user

So each Genie answer respects that user's permissions

Lineage and tags

Trace any KPI back to the raw drawing

Checked AI

In the app

Visible SQL

Every answer shows the query behind it

Grounding checks

Every number in the daily brief is checked against the data

Clause and evidence cited

Each flag cites the clause and shows the evidence text

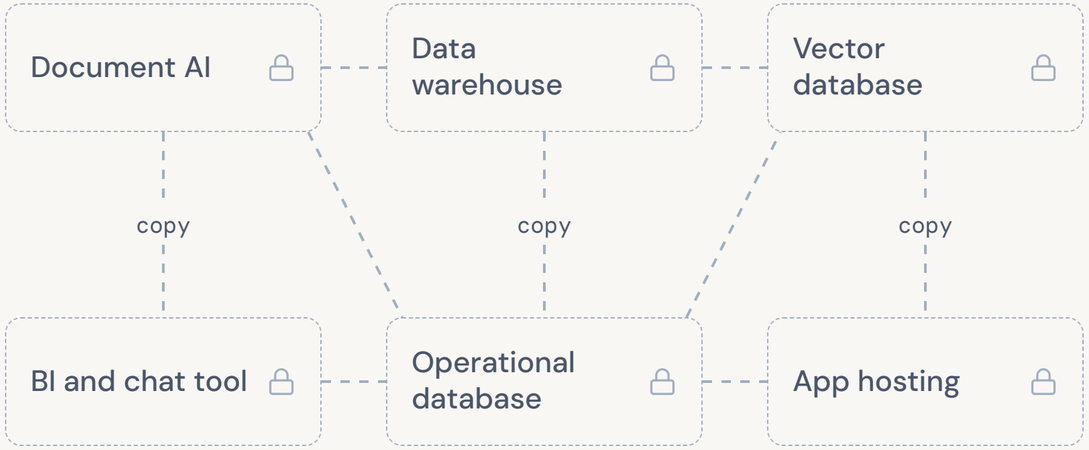
Labeled AI output

Engineers make the call ⚠ AI finding · verify before action

WHY DATABRICKS

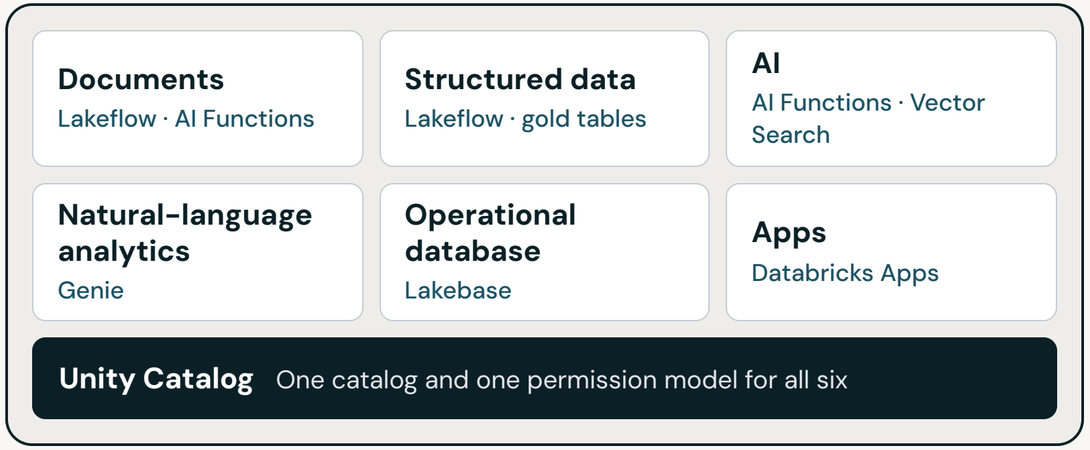
One platform for every piece, governed once

POINT SOLUTIONS STITCHED TOGETHER



- Data copied between vendors
- A separate permission model in every tool
- Lineage breaks at each hand-off

DATABRICKS DATA INTELLIGENCE PLATFORM



- Data stays in one place, with no copies across vendors
- One permission model for data, AI and apps
- Lineage end to end, from drawing to KPI

PATH FORWARD

A 6–8 week pilot on one live Bechtel project

SUGGESTED PILOT PLAN

Weeks 1–2

Set up

Data access; load the drawing set and standards

Weeks 3–5

Tune

Calibrate on Bechtel standards and historical NCRs with SMEs

Weeks 6–8

Use and measure

PMs and engineers use the Co-Pilot; joint readout

→ Week-8 go/no-go: scale across the portfolio if the measured case clears break-even

BECHTEL PROVIDES

- One live project's drawing set
- Applicable standards and checklists
- 2–3 subject-matter experts

SUCCESS MEASURES

- Recall on real violations
- Engineering review hours saved
- PM adoption
- Rework avoided vs. break-even

NEXT STEPS

1. **Pilot-selection workshop**
   Choose the project and agree the go/no-go criteria with the executive sponsor
2. **Data-access setup**
   Drawings, standards and project data in a governed workspace
3. **Kickoff date**
   Agree the start of the 6–8 week pilot

Book the pilot workshop →